

Pt100 series

Thin Film Platinum RTDs

H = 1.0±0.1mm (Height)
L = 2.3±0.1mm (Length)
W = 2.0±0.1mm (Width)
LW = 10±1mm (Lead length)
Lφ = 0.2±0.02mm (Lead diameter)

MAIN FEATURES

- PT100 series thin film platinum resistors have the advantages of small size, high precision and good long-term stability.
- It has the characteristics of anti-vibration and anti-shock.
- The product can be subdivided into regular ultra low and high temperature series, covering the temperature range of -200 to 650°C.
- It can be used in many connection ways, such as resistance welding, argon arc welding, pressure welding, brazing and so on.
- Widely used in automotive, instrumentation, household appliances, new energy and other fields.

*The nominal resistance measurement point is 8mm away from the component body



TECHNICAL INDEX

Performance parameters	Thin film platinum resistance series	Component size	2.0x2.3x1.0mm	1.6x2.0x1.0mm	1.2x2.0x1.0mm
Lead specifications	Length:10mm; Diameter:0.2mm	Lead material	Platinum nickel; Silver target; Pure Platinum; Sterling silver;		
Lead tension	≥9N	Insulation impedance	>100MΩ at 20℃、> 2MΩ at 500℃		
Temperature coefficient (TCR)	3850ppm/℃	Working current	0.3- 1mA (Self-heating should be considered)		
Long-term stability	After 1000 hours at 500℃ , the resistance shift of R(0℃) is less than0.04%				
Response time	Water flow(v=0.4m/s)	τ _{0.5} =0.1s	τ _{0.9} =0.3s	Airflow(v=2m/s)	τ _{0.5} =5s τ _{0.9} =15s
Natural coefficient	0℃ 0.4℃/mW	Anti-vibration	Frequency acceleration ≥40g from 10 to 2000Hz		
Impact resistant	8ms half sine wave acceleration ≥100g	Package	Vacuum plastic packaging (Provide other packaging forms as required)		
Others	Substrate size, basic resistance value, lead specifications (can be provided upon request)				

REFERENCE SELECTION

Type	Range of application	Classes	R ₀ (Ω)	Temperature range	Accuracy
Pt100-N500	-70~+500°C	1/10B	100±0.01	0~+100°C	±(0.03+0.0005 T)
		1/3B	100±0.04	0~+150°C	±(0.1+0.0017 T)
		A	100±0.06	-50~+300°C	±(0.15+0.002 T)
		B	100±0.12	-70~+500°C	±(0.3+0.005 T)
		2B	100±0.24	-70~+500°C	±(0.6+0.01 T)
Pt100-GW650	-70~+650°C	B	100±0.12	-70~+650°C	±(0.3+0.005 T)
		2B	100±0.24	-70~+650°C	±(0.6+0.01 T)
Pt100-DW200	-200~+150°C	B	100±0.12	-200~+150°C	±(0.3+0.005 T)
		2B	100±0.24	-200~+150°C	±(0.6+0.01 T)

Note * : the marked classes and temperature measurement accuracy refer to the IEC60751 standard. T is the measured temperature.